

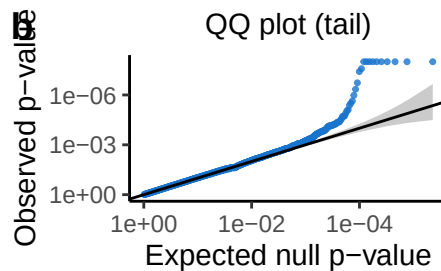
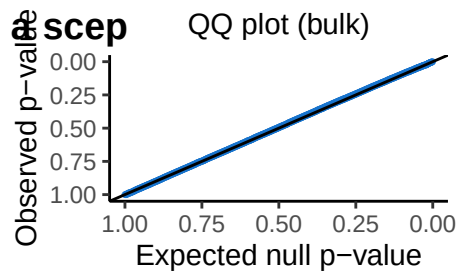
reprogle-analysis-highly-expressed

2026-03-19

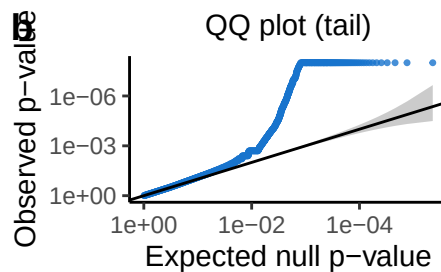
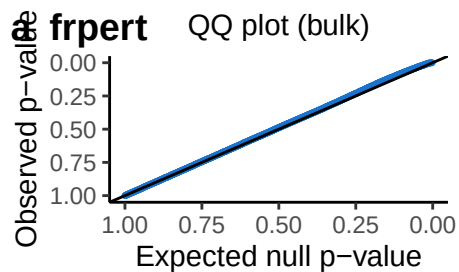
This notebook makes figures to explore the following:

analysis of positive and negative control on reprogle, only using highly expressed genes.

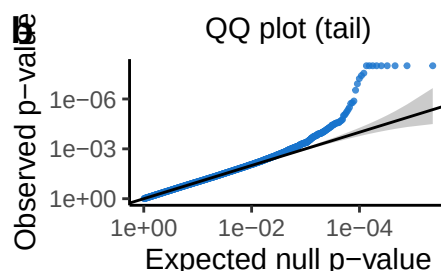
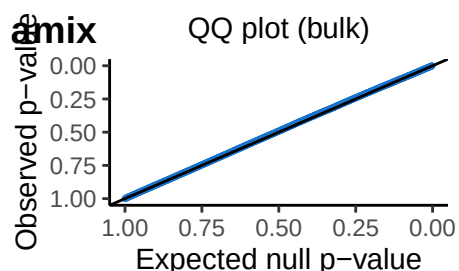
```
## Dataset: reprogle-rd7_neg_control_4000genes : #genes = ; #cells = 13548
## #guides = 113 ; #targets = 113 ; avg # guides / target: 1
## # cells per guide: median = 111 ; min = 29 ; max = 617
## # cells per target: avg = 111 ; min = 29 ; max = 617
## # guides per cell: moi (avg) = 1 ; min = 1 ; max = 1
## # discovery pairs: 293348
##
## *****
## Dataset: reprogle-rd7_pos_control_large : #genes = 1651 ; #cells = 210084
## #guides = 1854 ; #targets = 1652 ; avg # guides / target: 1.12
## # cells per guide: median = 86 ; min = 8 ; max = 1766
## # cells per target: avg = 87 ; min = 20 ; max = 10058
## # guides per cell: moi (avg) = 1 ; min = 1 ; max = 1
## In total: 1651 pos. ctrl pairs and 293348 neg. ctrl pairs.
##      pass_qc_75 pass_qc_50 pass_qc_30 pass_qc_20 pass_qc_trt pass_qc_def
## FALSE      179170      64578      5452      1032      119344      175240
## TRUE       115829     230421     289547     293967     175655     119759
```



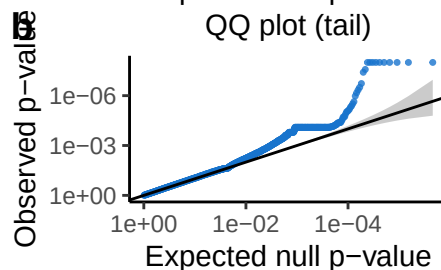
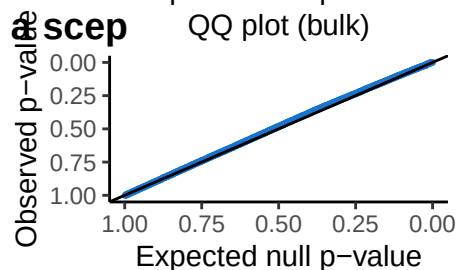
QC: pass_qc_75
Num. signif
(at alpha 0.1):
24 of 114356



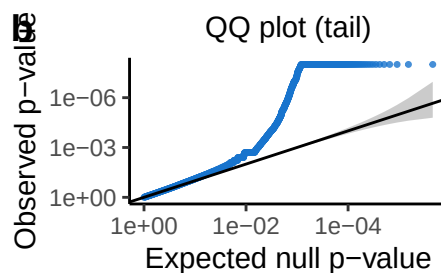
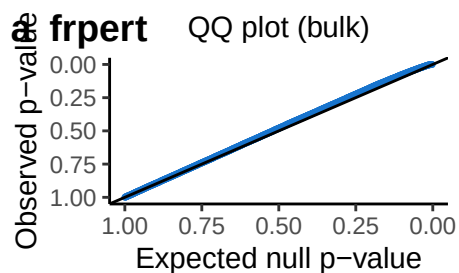
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697 of 114356



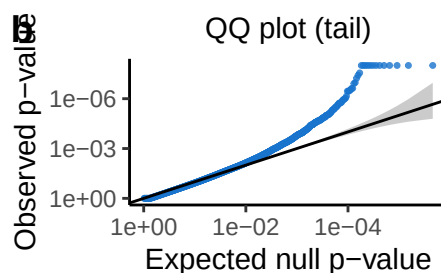
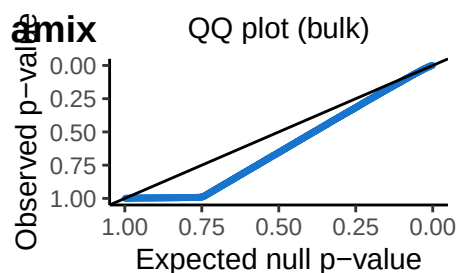
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27 of 114356



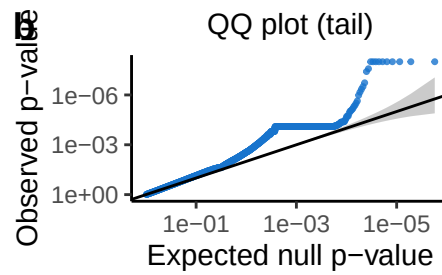
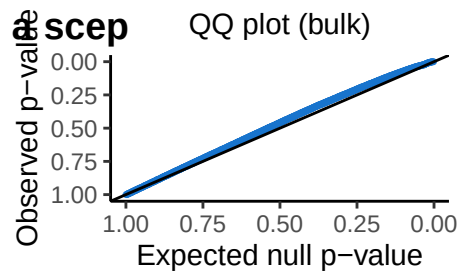
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Num. signif
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258 of 228825



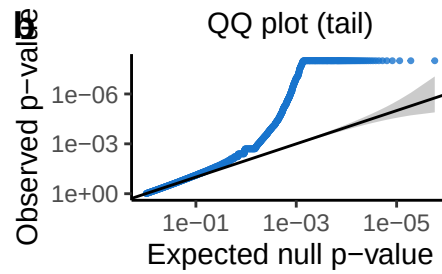
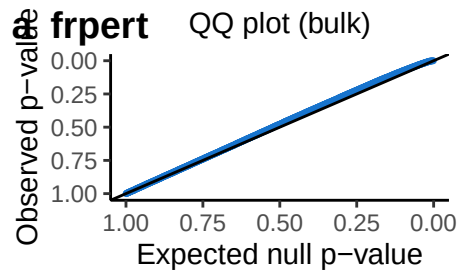
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(at alpha 0.1):
1096 of 228825



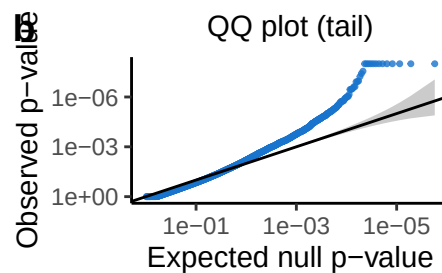
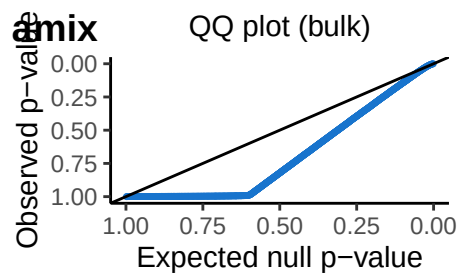
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138 of 228825



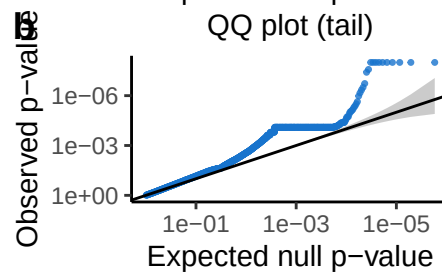
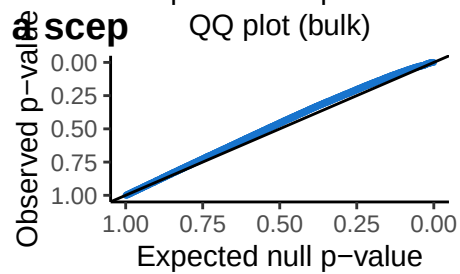
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1283 of 287924



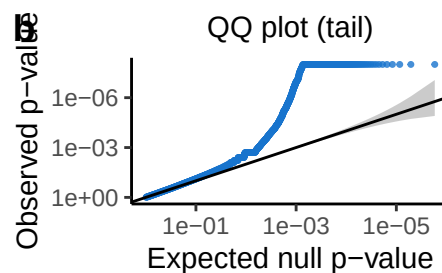
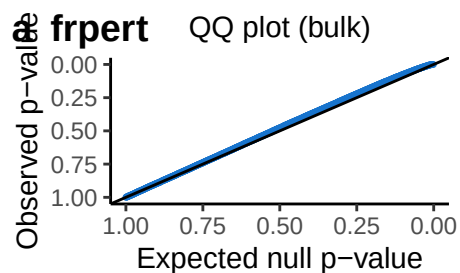
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1305 of 287924



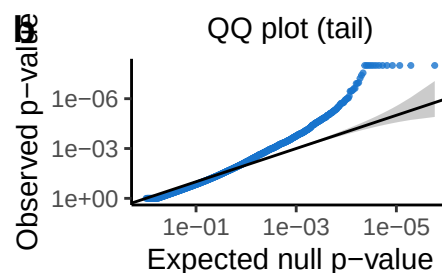
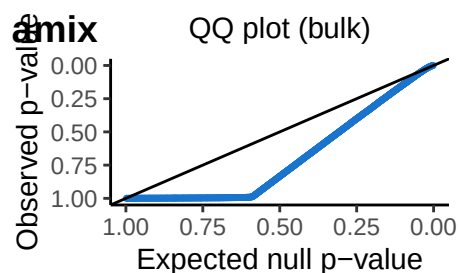
QC: pass_qc_30
Num. signif
(at alpha 0.1):
152 of 287916



QC: pass_qc_20
Num. signif
(at alpha 0.1):
1315 of 292331



QC: pass_qc_20
Num. signif
(at alpha 0.1):
1314 of 292331



QC: pass_qc_20
Num. signif
(at alpha 0.1):
152 of 292323

Separation of pos. and neg. (QC = pass_qc_75)

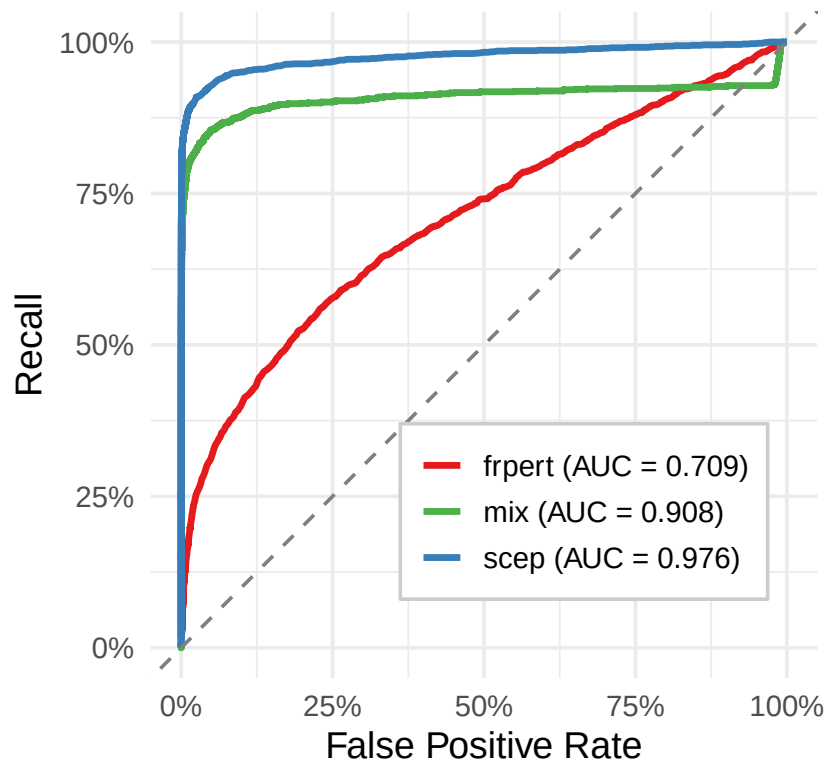
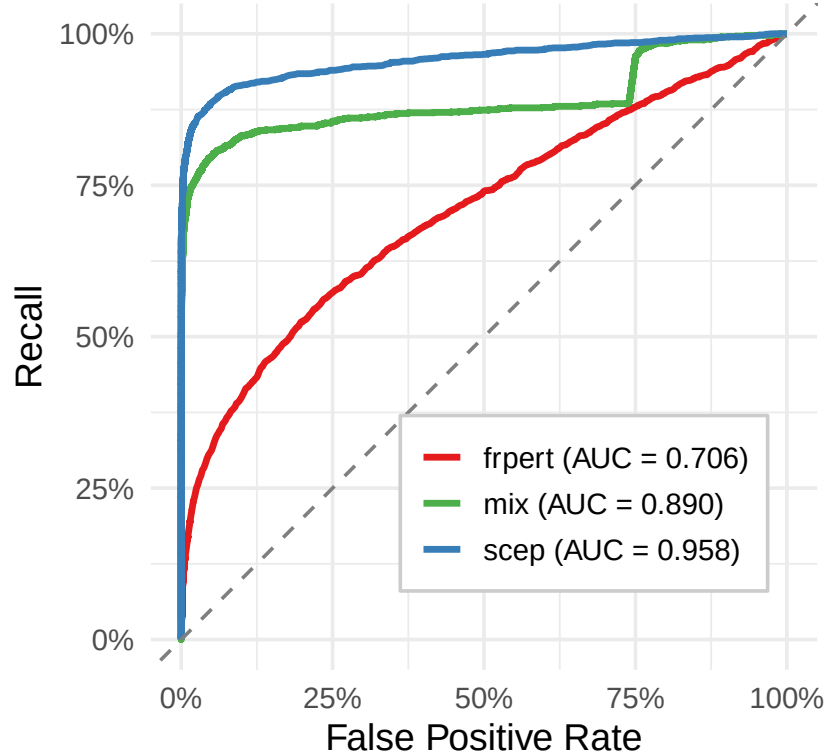
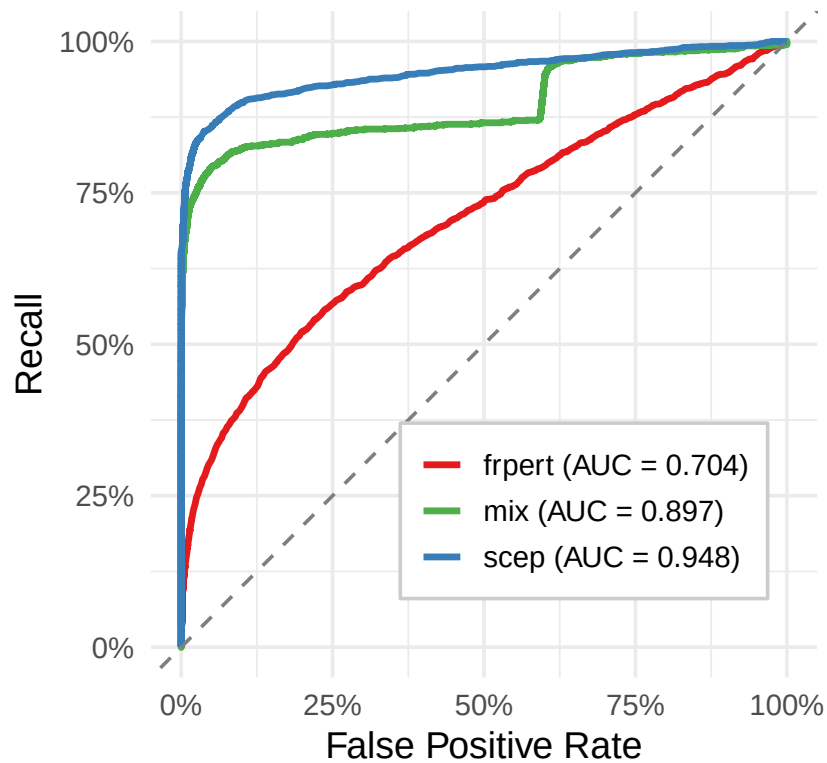


Figure 1: ROC analysis

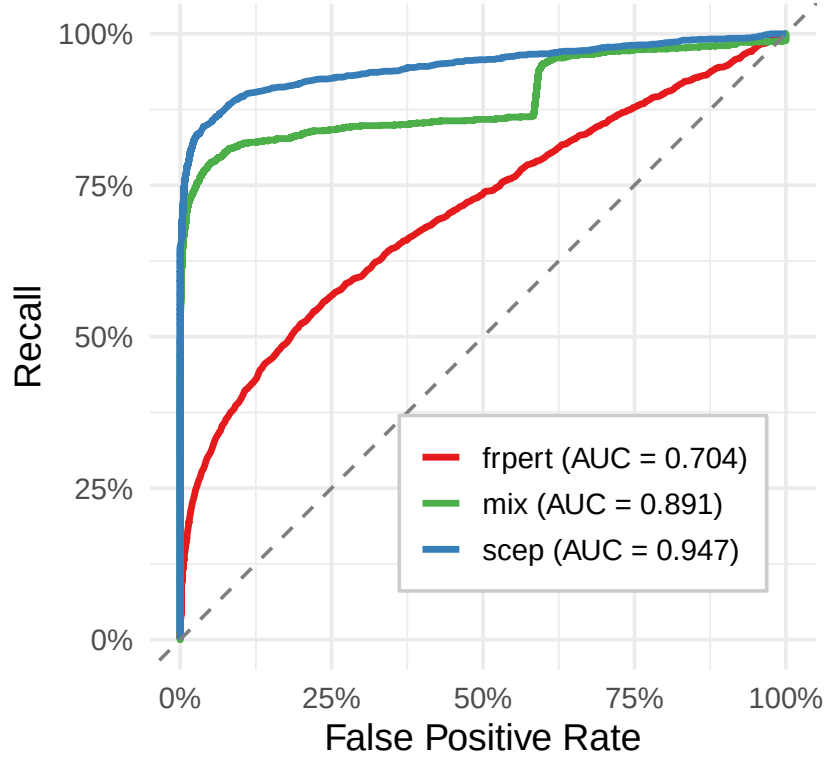
Separation of pos. and neg. (QC = pass_qc_50)



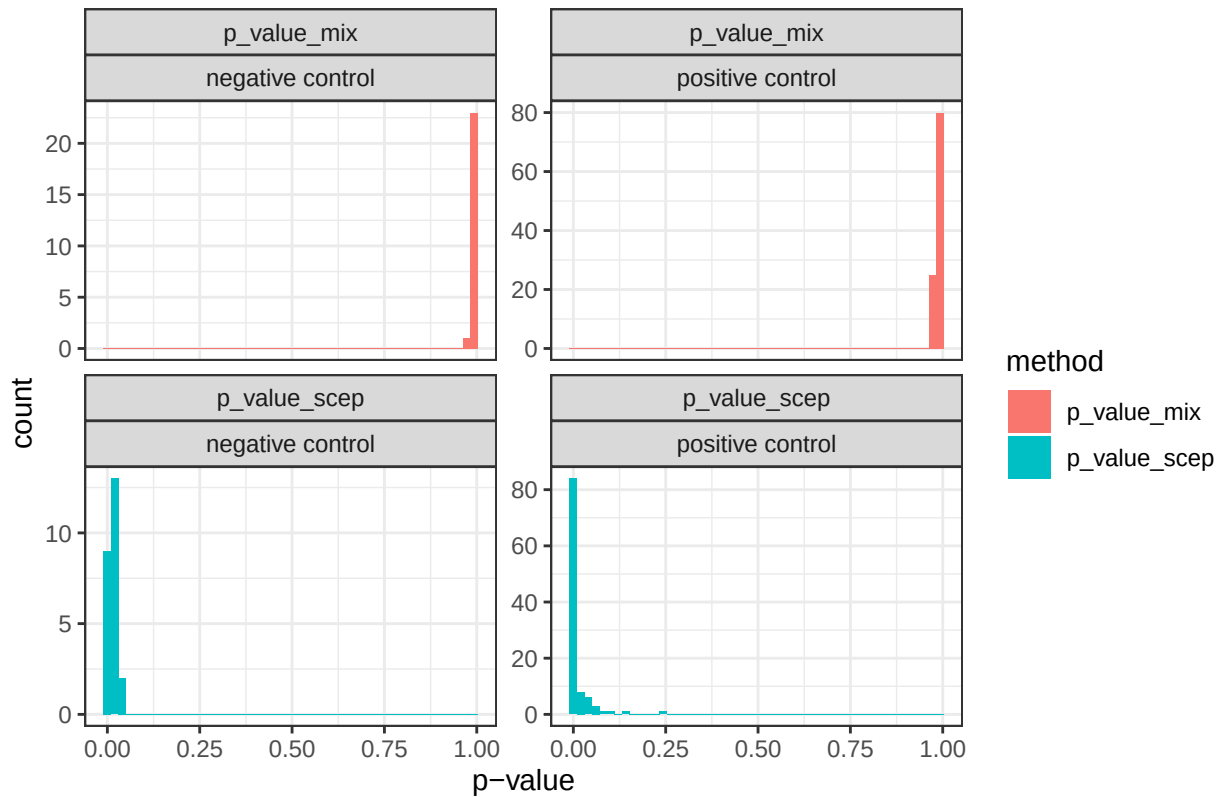
Separation of pos. and neg. (QC = pass_qc_30)



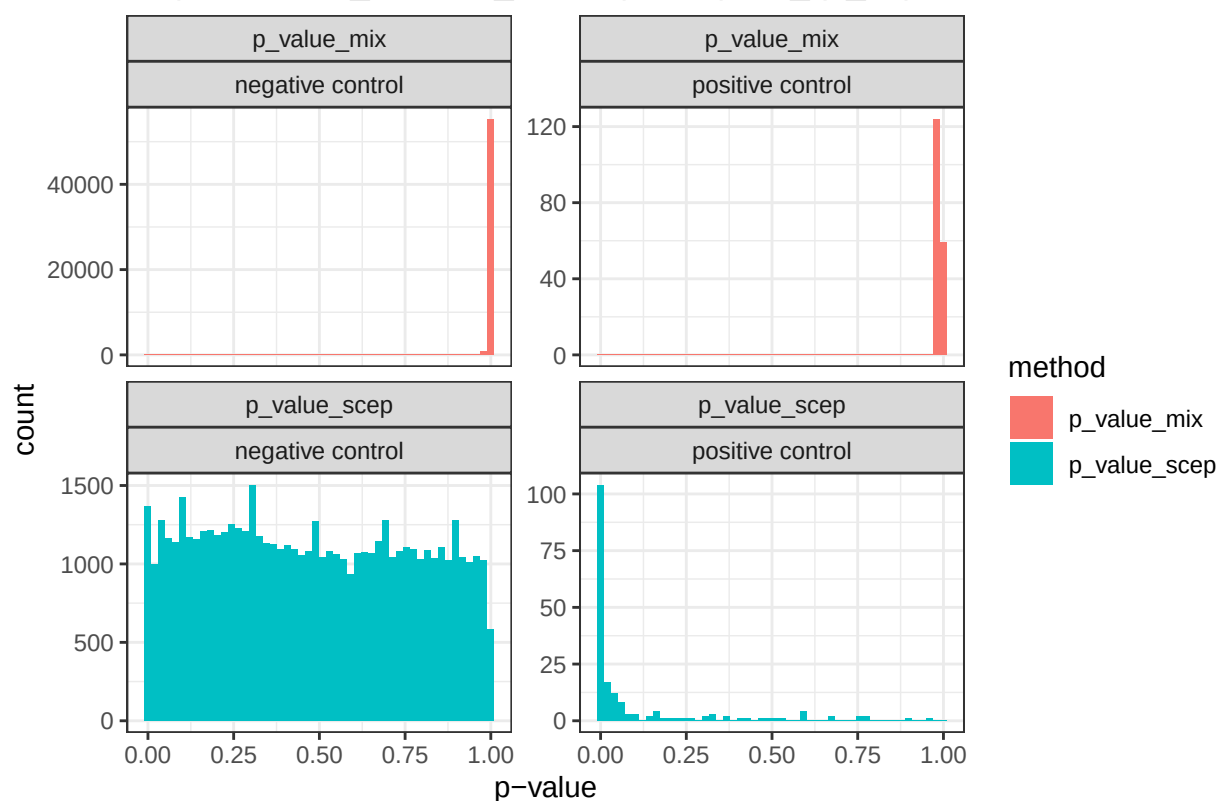
Separation of pos. and neg. (QC = pass_qc_20)



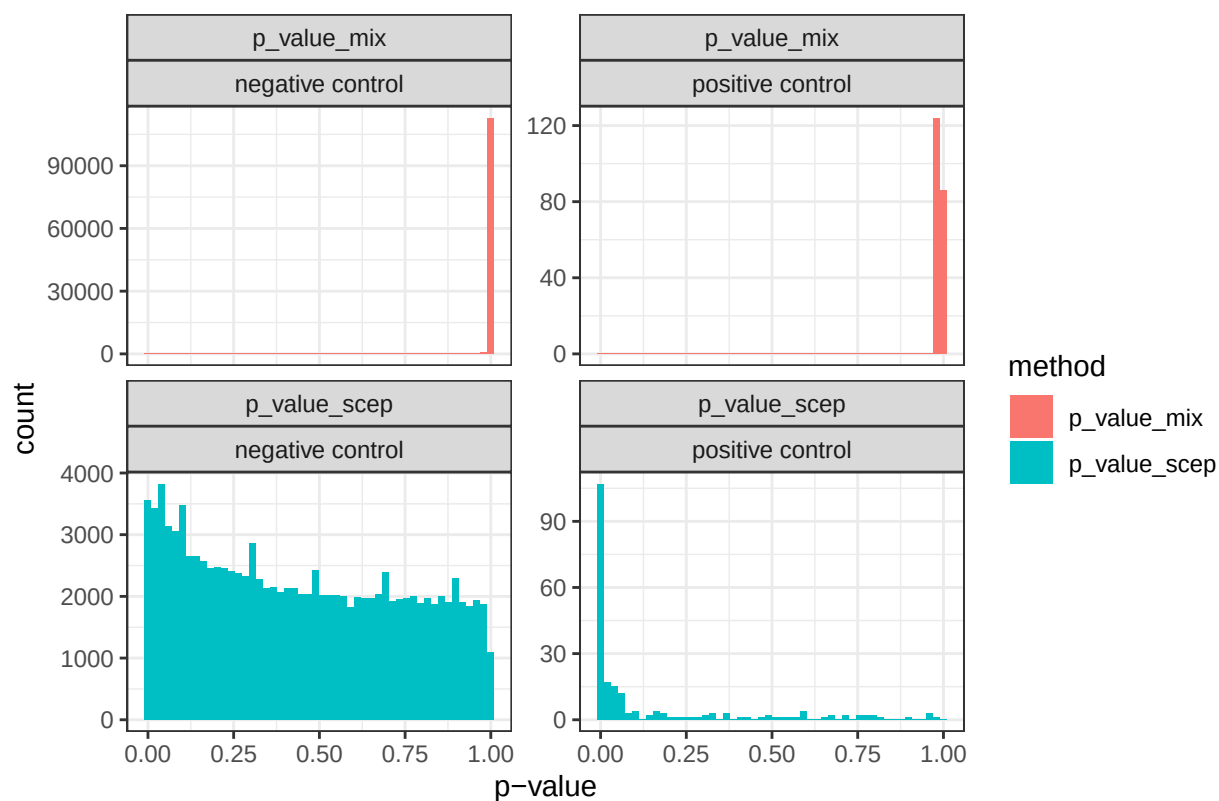
All pairs with n_nonzero_trt = 0 (QC = pass_qc_75)



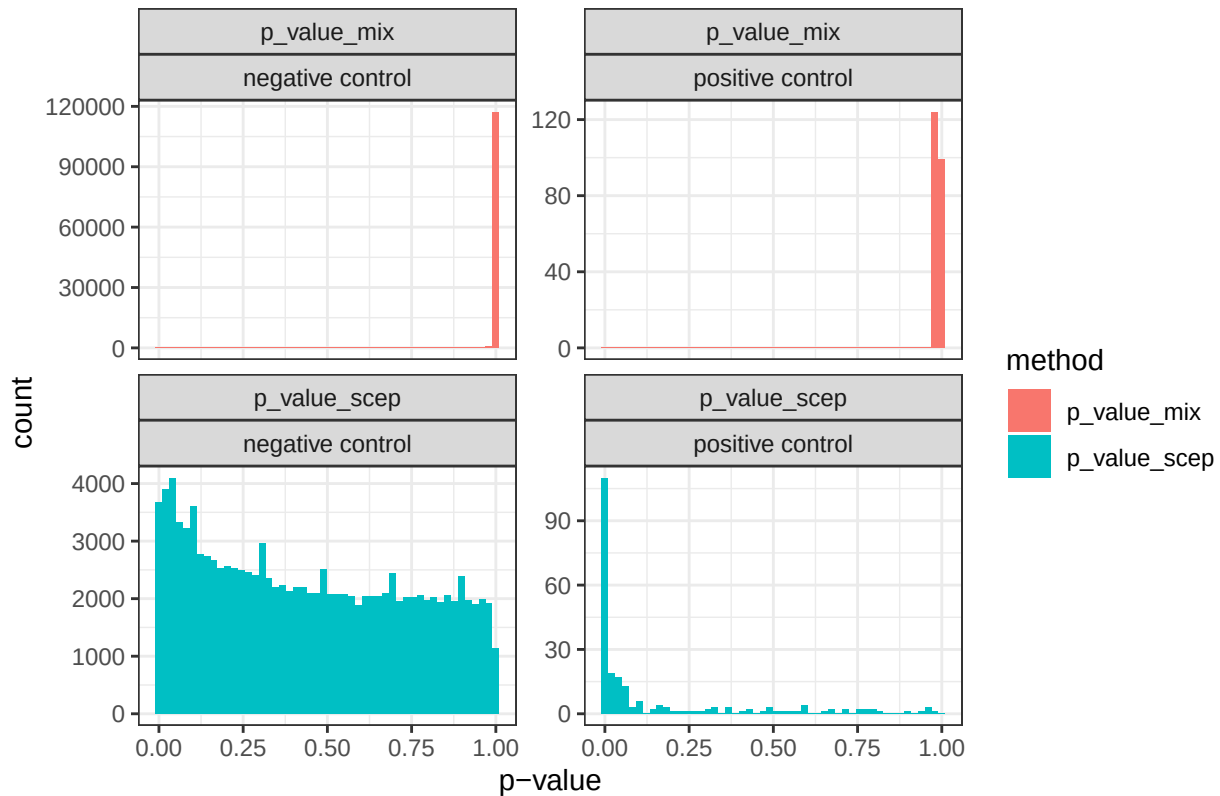
All pairs with $n_nonzero_trt = 0$ (QC = pass_qc_50)



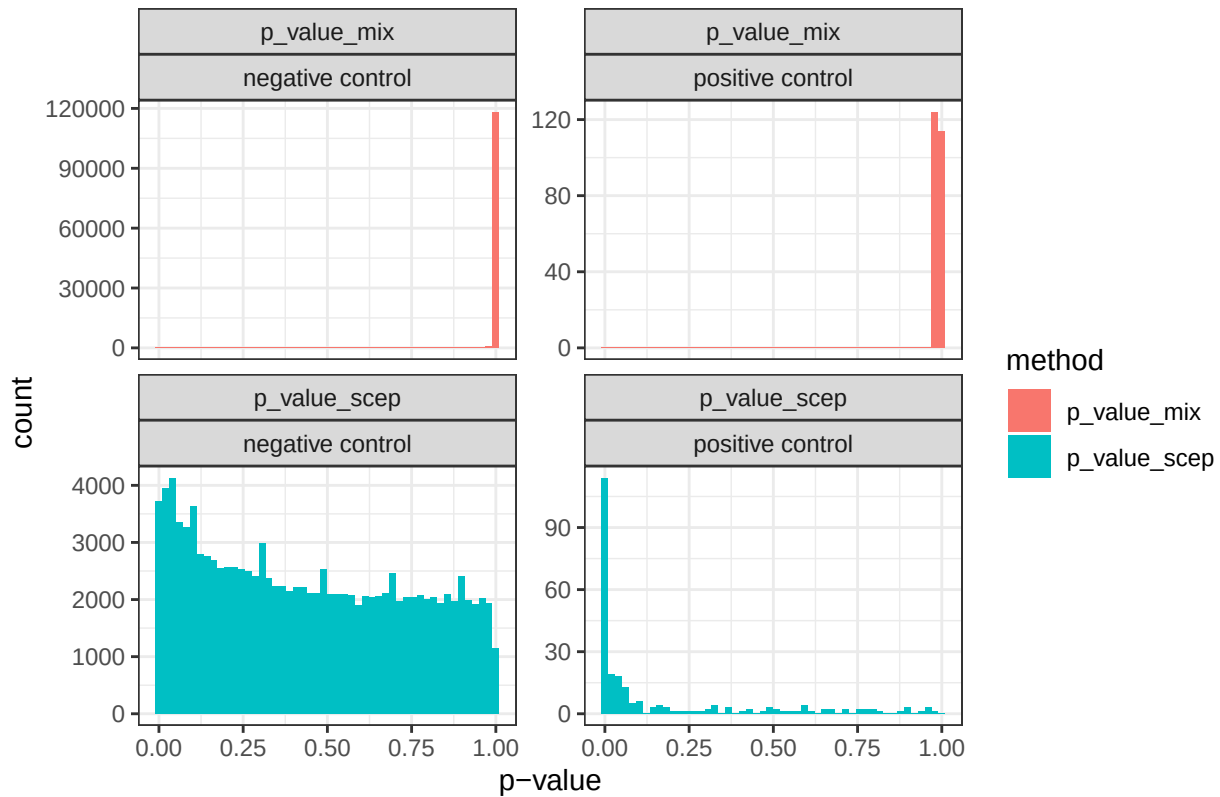
All pairs with $n_nonzero_trt = 0$ (QC = pass_qc_30)



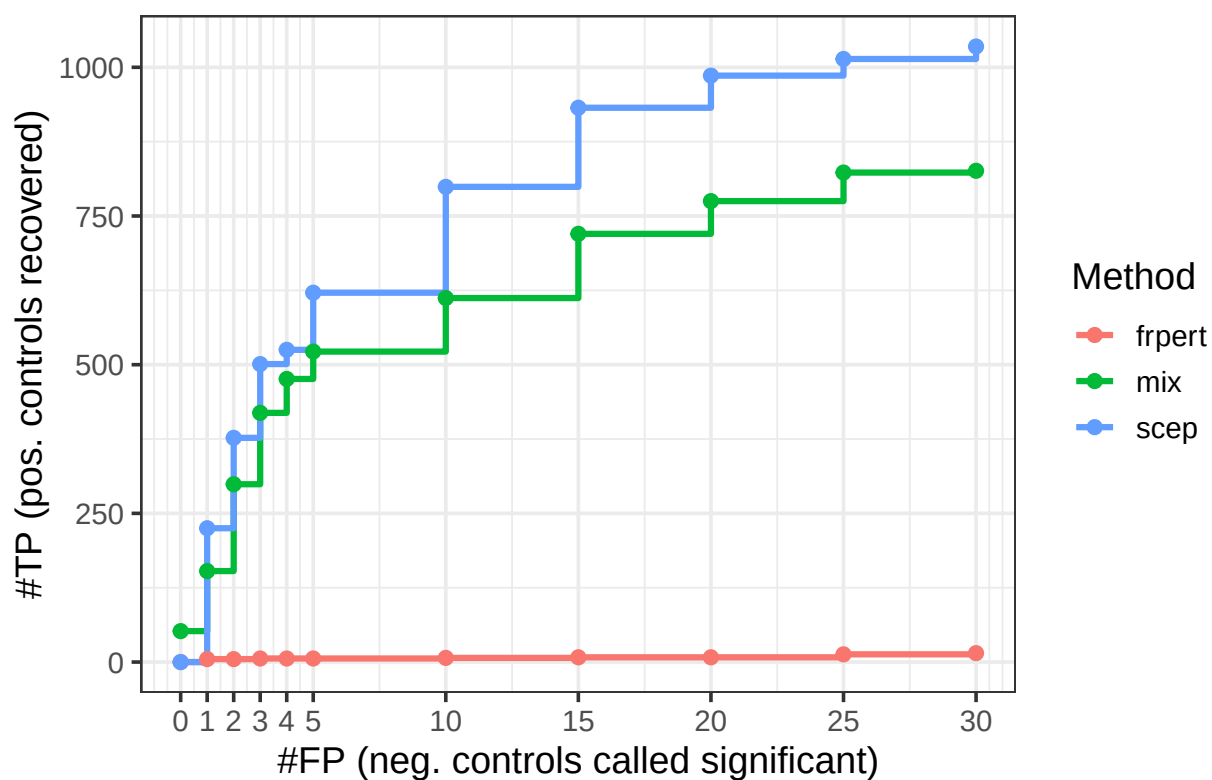
All pairs with n_nonzero_trt = 0 (QC = pass_qc_20)



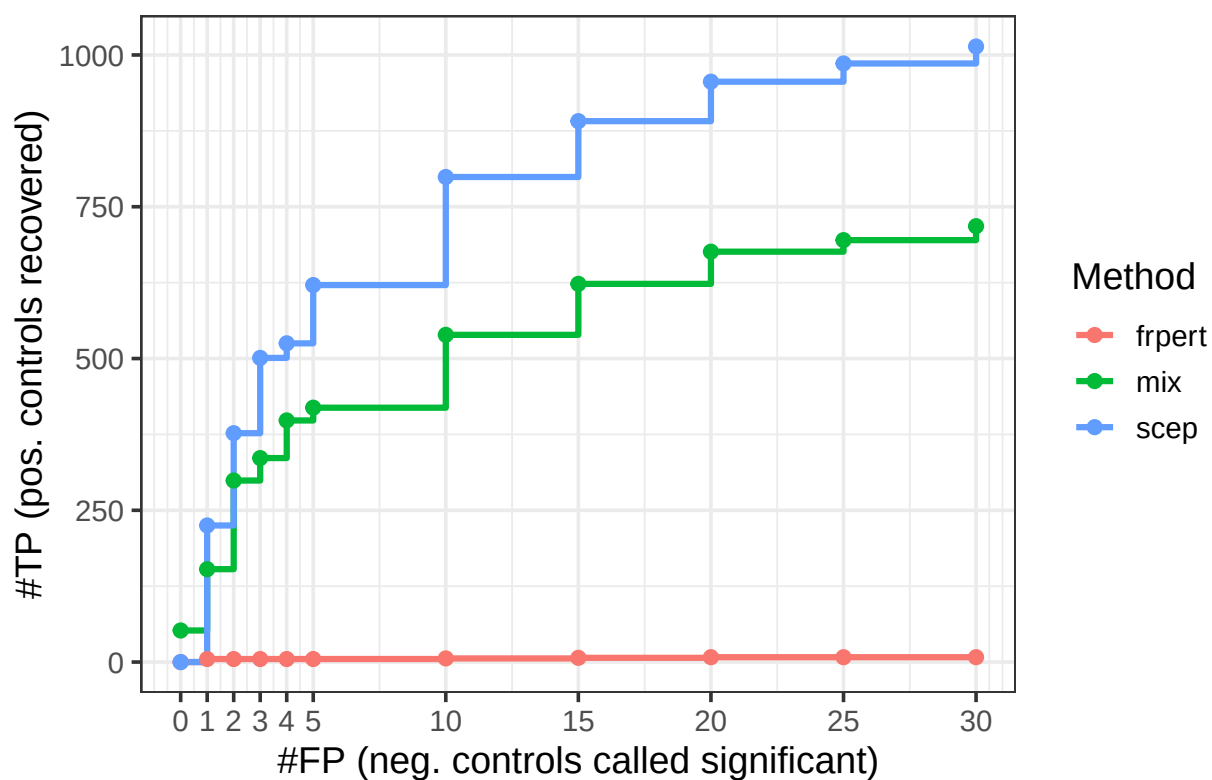
All pairs with n_nonzero_trt = 0 (no QC)



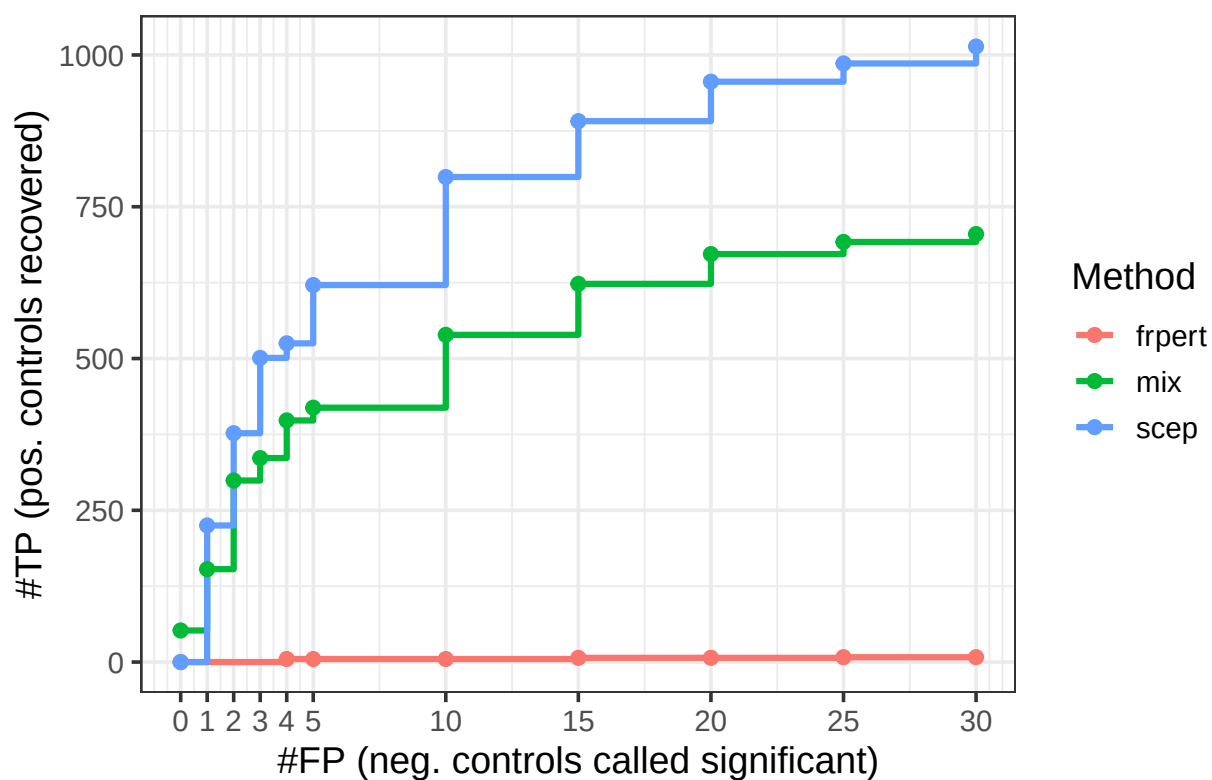
#TP recovered vs #FP allowed (QC = pass_qc_75)



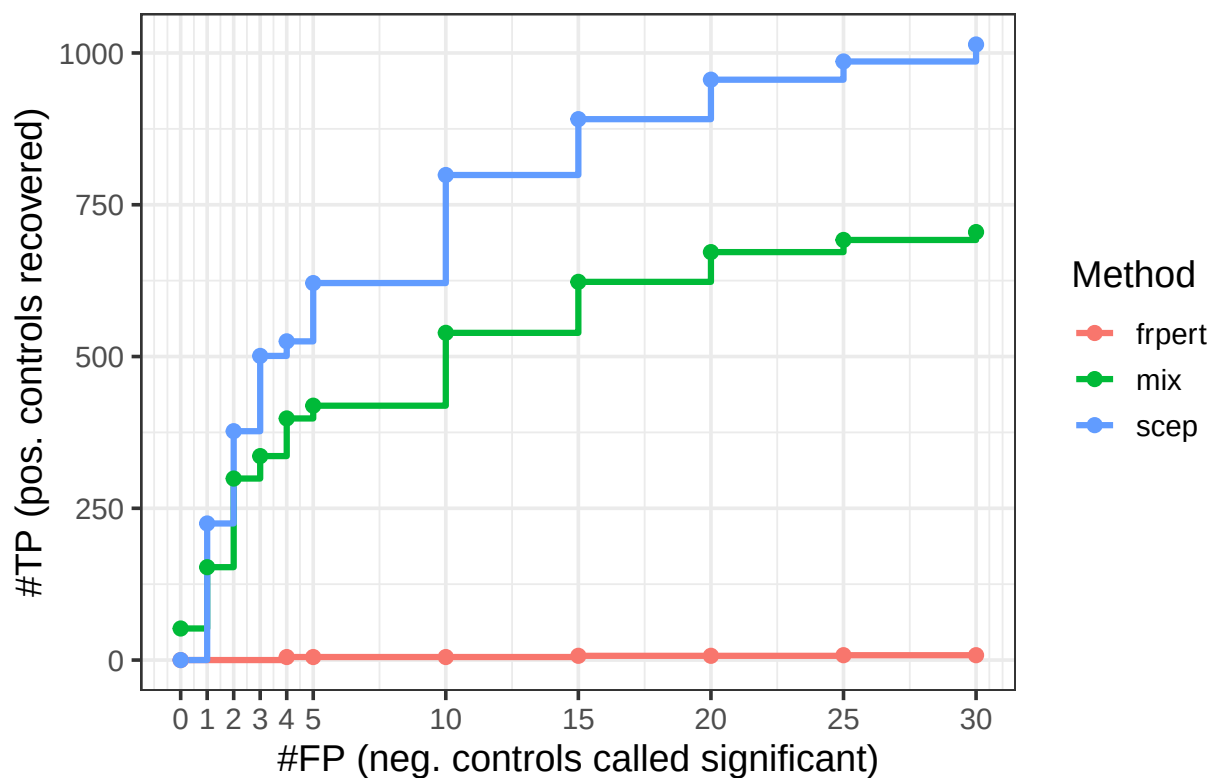
#TP recovered vs #FP allowed (QC = pass_qc_50)



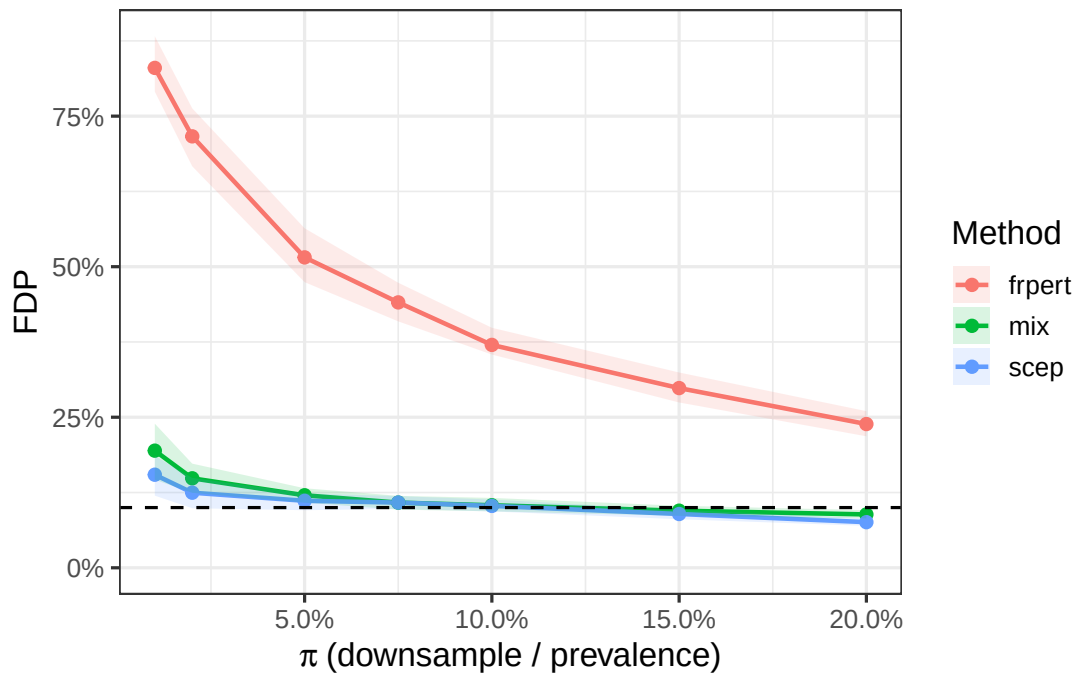
#TP recovered vs #FP allowed (QC = pass_qc_30)



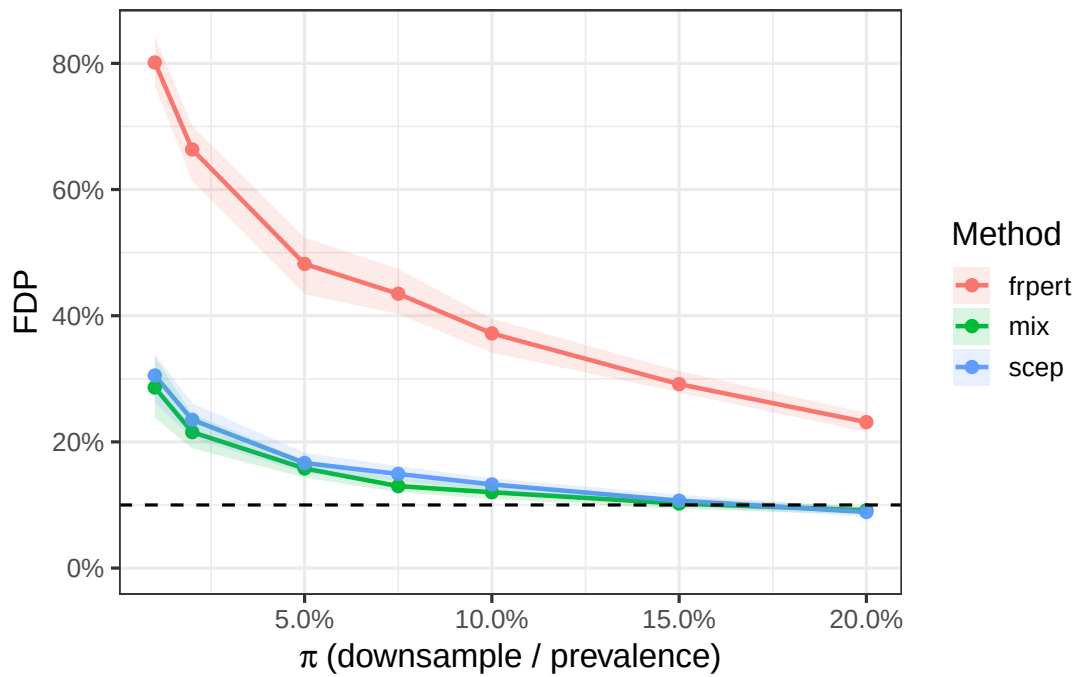
#TP recovered vs #FP allowed (QC = pass_qc_20)



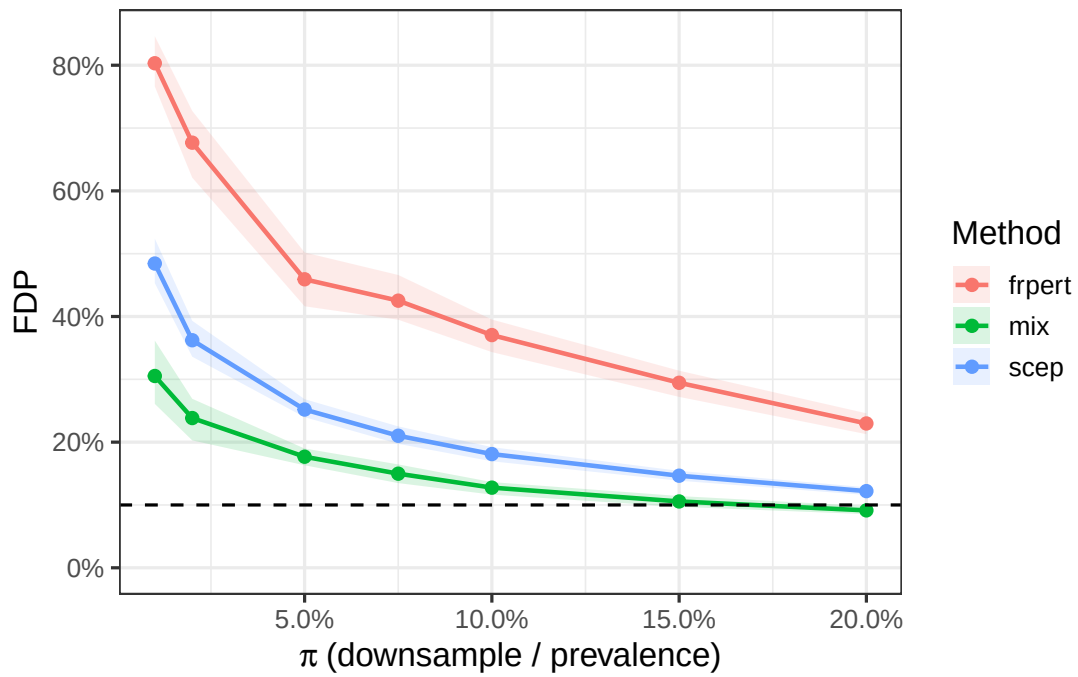
BH $q = 0.1$ (N=5000, B=100, QC = pass_qc_75)



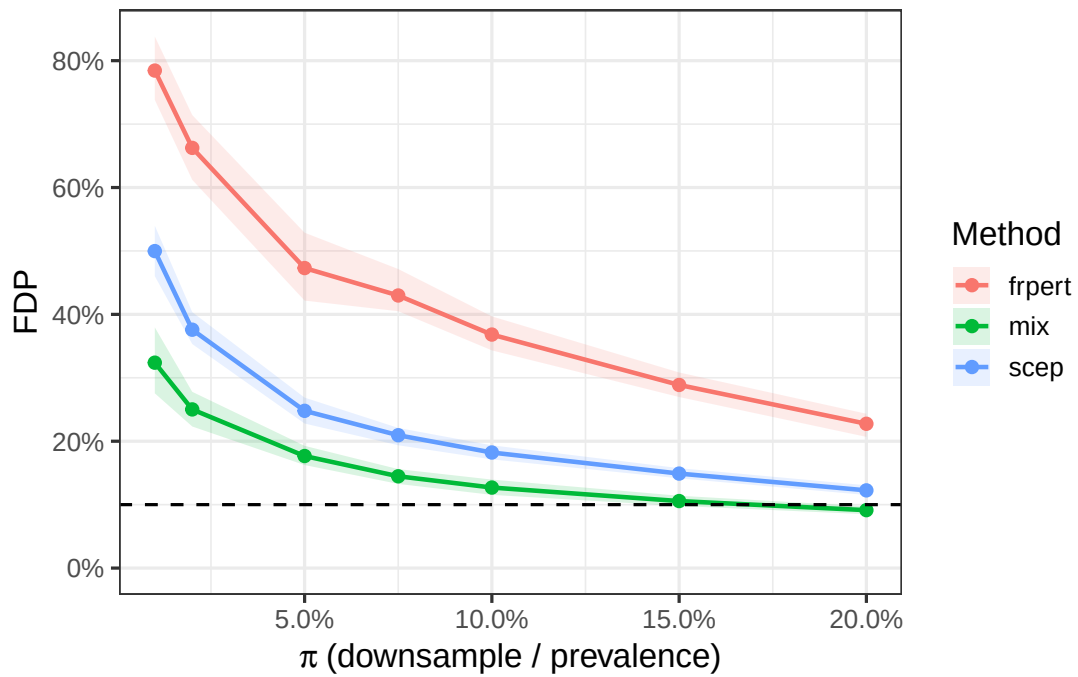
BH $q = 0.1$ (N=5000, B=100, QC = pass_qc_50)



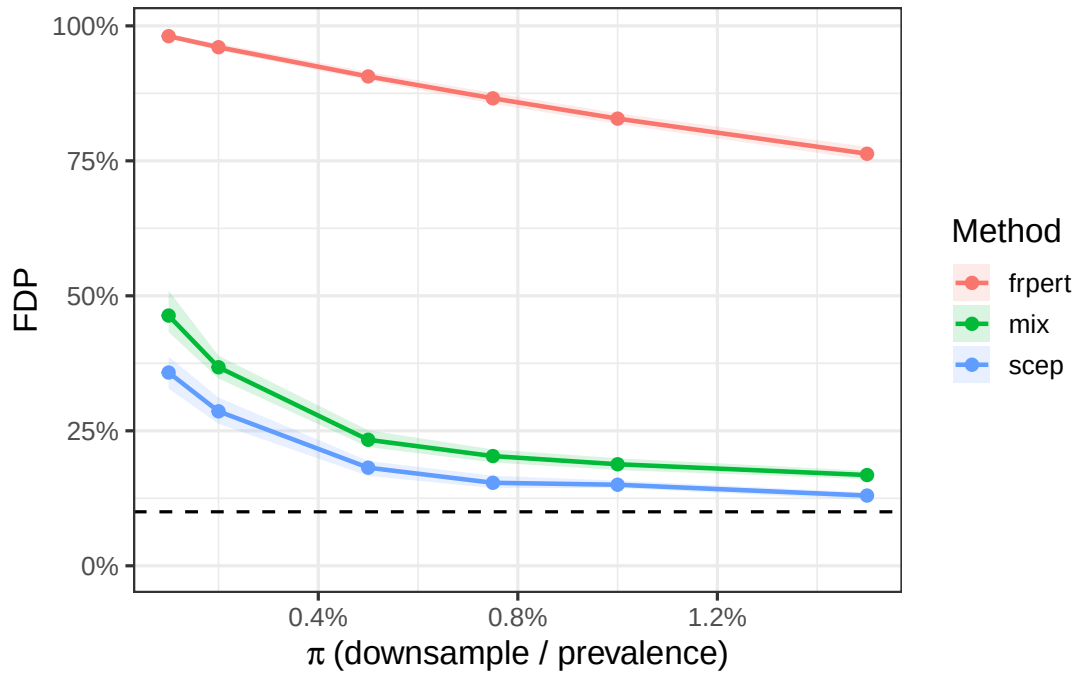
BH $q = 0.1$ (N=5000, B=100, QC = pass_qc_30)



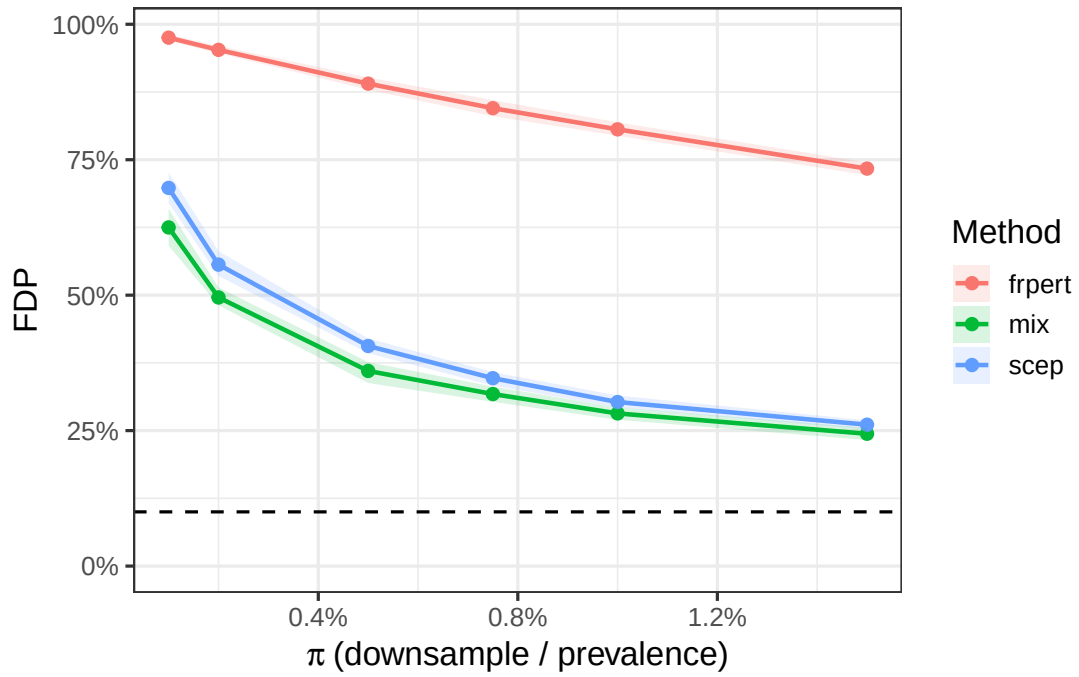
BH $q = 0.1$ (N=5000, B=100, QC = pass_qc_20)



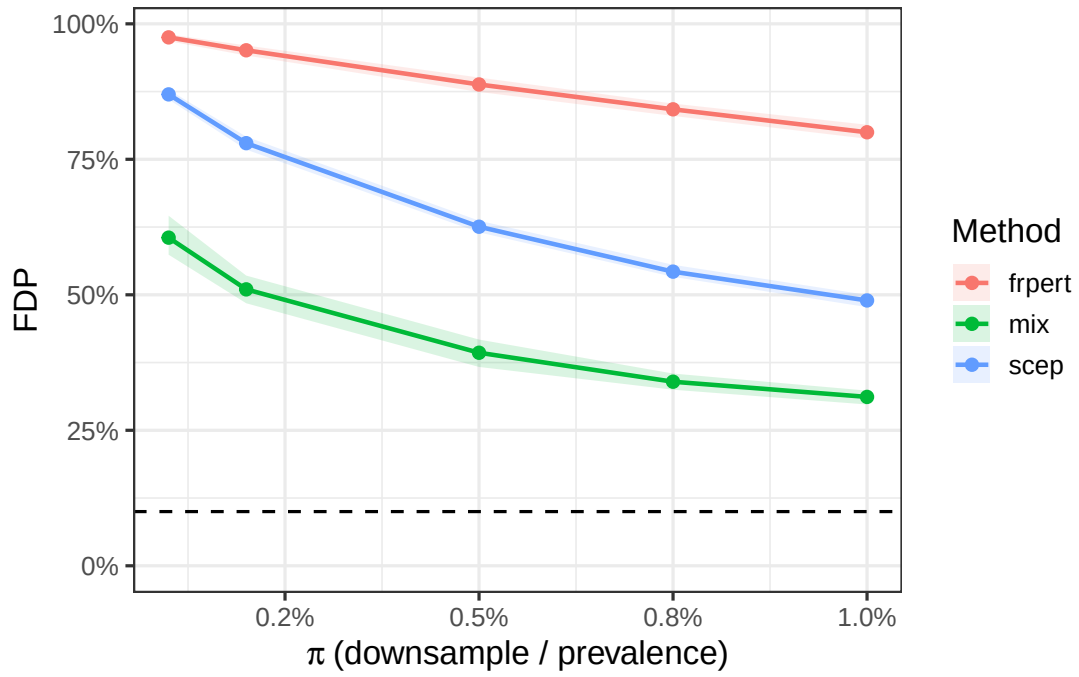
BH $q = 0.1$ (N=50000, B=100, QC = pass_qc_75)



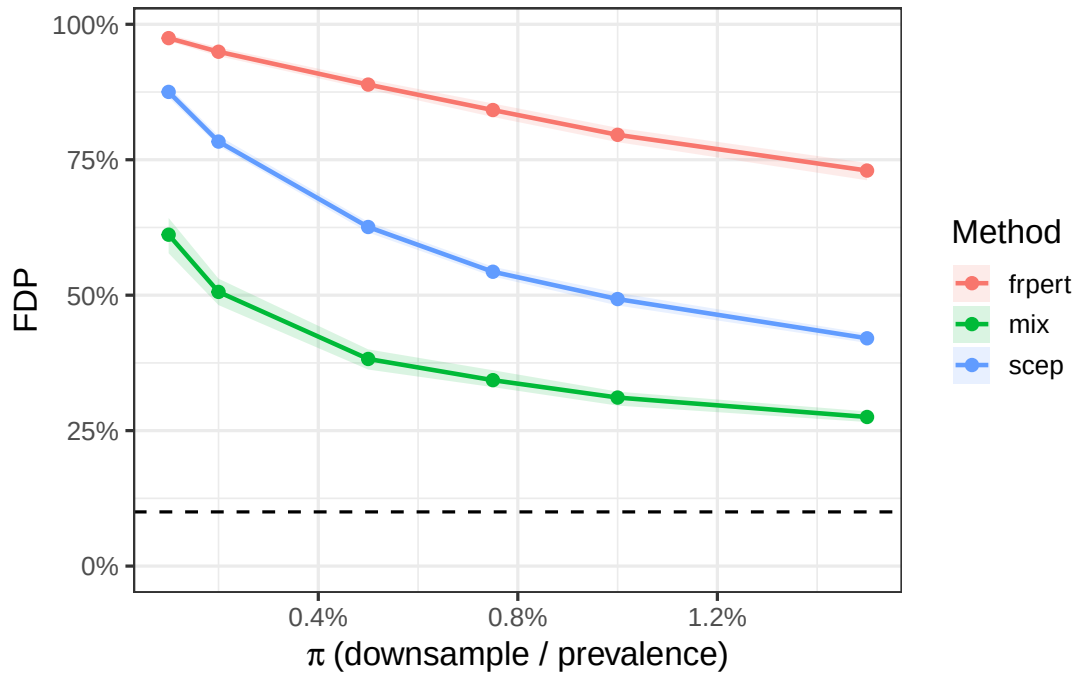
BH $q = 0.1$ (N=50000, B=100, QC = pass_qc_50)



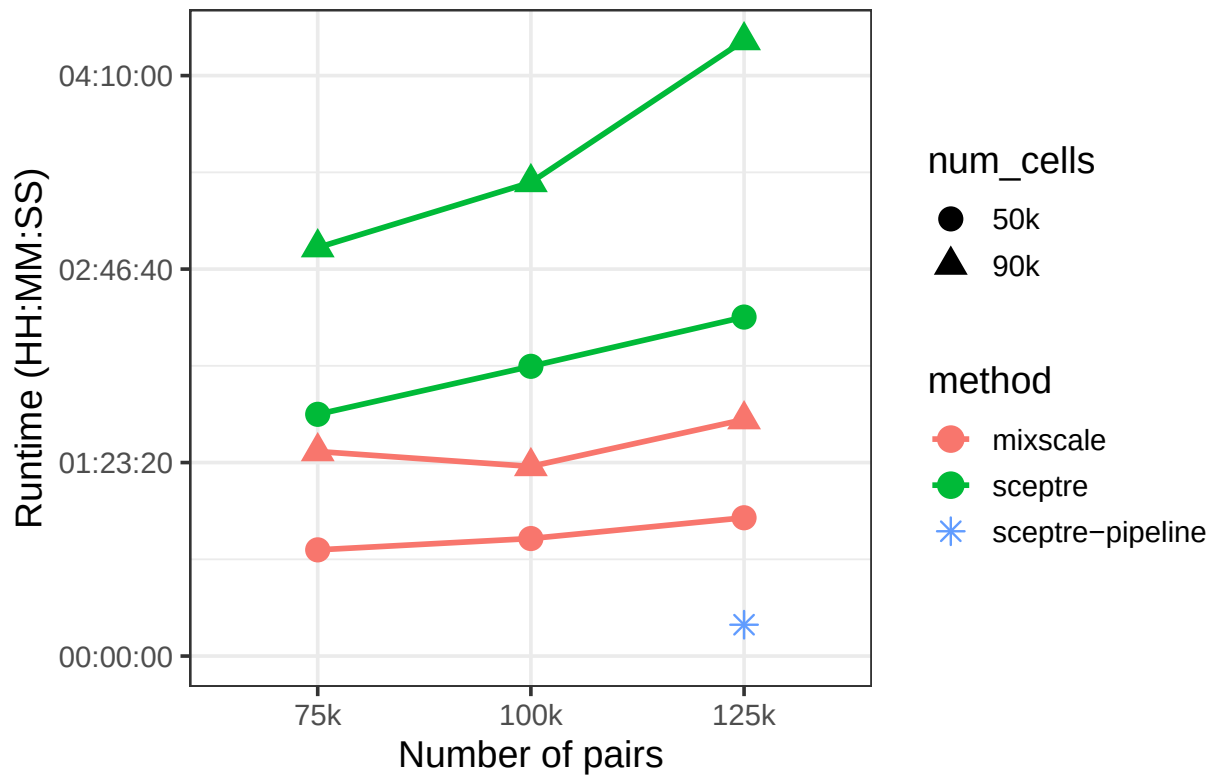
BH $q = 0.1$ (N=50000, B=100, QC = pass_qc_30)



BH $q = 0.1$ (N=50000, B=100, QC = pass_qc_20)



Runtime for highly expressed genes



Memory for highly expressed genes

